FACTORING A MAP FROM A UNIFORM MODULE THROUGH A SUFFICIENTLY NEGATIVE SHIFT OF THE IDEAL POWERS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The notion of an (A, n, c)-module and the compatible system of canonical maps from the pullback of the ideal powers to the structure sheaves of the closed subschemes. Set up in the author's section on a bit of uniformity, where the counted statement occurs, and used at the start of the counted proof. Fixes the hypothesis on the module and the maps along which the given homomorphism is pulled back.
- Duality for the module and the description of maps out of a near-projective module. Both are cross-chapter pointers invoked inside the counted proof. Convert the pullback of the given map into a map of dual modules, which is where the shift in the ideal power comes from.

Source reference. Original lemma lemma-annihilated: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5241–5241. Navigation pointer only: the counted text refers to the Stacks item lemma-annihilated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-bound-q-and-d: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5200–5200. Navigation pointer only: the counted text refers to the Stacks item lemma-bound-q-and-d, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-vanishing-coh-almost-projective: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5373–5373. Navigation

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Source reference. Original reference more-algebra-lemma-dual-near-projective: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5375–5375. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-dual-near-projective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original remark remark-duals: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5361–5361. Navigation pointer only: the counted text refers to the Stacks item remark-duals, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *With $q_0 = q(S)$ and $d = d(S)$ as above, let M be an (A, n, c)-module and let $\varphi : M \to I^n/I^{2n}$ be an A-linear map. Assume $n \geq \max(q_0+(1+d)c, (2+d)c)$ and if $d = 0$ assume $n \geq q_0 + 2c$. Then the composition*

$$M \xrightarrow{\varphi} I^n/I^{2n} \to I^{n-(1+d)c}/I^{2n-(1+d)c}$$

is of the form $\sum a_i\psi_i$ with $a_i \in I^c$ and $\psi_i : M \to I^{n-(2+d)c}/I^{2n-(2+d)c}$.

Proof. The case $d > 1$. Since we have a compatible system of maps $p^*(I^q) \to \mathcal{O}_X(q)$ for $q \geq 0$ there are canonical maps $p^*(I^q/I^{q+\nu}) \to \mathcal{O}_{Y_\nu}(q)$ for $\nu \geq 0$. Using this and pulling back φ we obtain a map

$$\chi : p^*M \longrightarrow \mathcal{O}_{Y_n}(n)$$

such that the composition $M \to H^0(X, p^*M) \to H^0(X, \mathcal{O}_{Y_n}(n))$ is the given homomorphism φ combined with the map $I^n/I^{2n} \to H^0(X, \mathcal{O}_{Y_n}(n))$. Since $\mathcal{O}_{Y_n}(n)$ is invertible on Y_n the linear map χ determines a section

$$\sigma \in \Gamma(X, (p^*M)^\vee(n))$$

with notation as in Remark remark-duals. The discussion in Remark remark-duals shows the cokernel and kernel of $can : p^*(M^\vee) \to (p^*M)^\vee$ are scheme theoretically supported on Y_c. By Lemma lemma-annihilated the map $(p^*M)^\vee(n) \to (p^*M)^\vee(n-2c)$ factors through $p^*(M^\vee)(n-2c)$; small detail omitted. Hence the image of σ in $\Gamma(X, (p^*M)^\vee(n-2c))$ comes from an element

$$\sigma' \in \Gamma(X, p^*(M^\vee)(n-2c))$$

By Lemma lemma-vanishing-coh-almost-projective part (5), the fact that $M^\vee$ is an (A, n, c)-module by More on Algebra, Lemma more-algebra-lemma-dual-near-projective, and the fact that $n \geq q_0 + (1+d)c$ so $n - 2c \geq q_0 + (d-1)c$ we see that the image of σ' in $H^0(X, p^*M^\vee(n-(1+d)c))$ is the image of an element τ in $I^{n-(1+d)c}M^\vee$. Write $\tau = \sum a_i\tau_i$ with $\tau_i \in I^{n-(2+d)c}M^\vee$; this makes sense as $n-(2+d)c \geq 0$. Then τ_i determines a homomorphism of modules $\psi_i : M \to I^{n-(2+d)c}/I^{2n-(2+d)c}$ using the evaluation map $M \otimes M^\vee \to A/I^n$.

Let us prove that this works[1]. Pick $z \in M$ and let us show that $\varphi(z)$ and $\sum a_i\psi_i(z)$ have the same image in $I^{n-(1+d)c}/I^{2n-(1+d)c}$. First, the element z determines a map

[1]We hope some reader will suggest a less dirty proof of this fact.

$p^*z : \mathcal{O}_{Y_n} \to p^*M$ whose composition with χ is equal to the map $\mathcal{O}_{Y_n} \to \mathcal{O}_{Y_n}(n)$ corresponding to $\varphi(z)$ via the map $I^n/I^{2n} \to \Gamma(\mathcal{O}_{Y_n}(n))$. Next z and p^*z determine evaluation maps $e_z : M^\vee \to A/I^n$ and $e_{p^*z} : (p^*M)^\vee \to \mathcal{O}_{Y_n}$. Since $\chi(p^*z)$ is the section corresponding to $\varphi(z)$ we see that $e_{p^*z}(\sigma)$ is the section corresponding to $\varphi(z)$. Here and below we abuse notation: for a map $a : \mathcal{F} \to \mathcal{G}$ of modules on X we also denote $a : \mathcal{F}(t) \to \mathcal{F}(t)$ the corresponding map of twisted modules. The diagram

$$\begin{array}{ccc} p^*(M^\vee) & \xrightarrow[p^*e_z]{} & \mathcal{O}_{Y_n} \\ {\scriptstyle can}\downarrow & & \| \\ (p^*M)^\vee & \xrightarrow{e_{p^*z}} & \mathcal{O}_{Y_n} \end{array}$$

commutes by functoriality of the construction *can*. Hence $(p^*e_z)(\sigma')$ in $\Gamma(Y_n, \mathcal{O}_{Y_n}(n-2c))$ is the section corresponding to the image of $\varphi(z)$ in I^{n-2c}/I^{2n-2c}. The next step is that σ' maps to the image of $\sum a_i\tau_i$ in $H^0(X, p^*M^\vee(n-(1+d)c))$. This implies that $(p^*e_z)(\sum a_i\tau_i) = \sum a_i p^*e_z(\tau_i)$ in $\Gamma(Y_n, \mathcal{O}_{Y_n}(n-(1+d)c))$ is the section corresponding to the image of $\varphi(z)$ in $I^{n-(1+d)c}/I^{2n-(1+d)c}$. Recall that ψ_i is defined from τ_i using an evaluation map. Hence if we denote

$$\chi_i : p^*M \longrightarrow \mathcal{O}_{Y_n}(n-(2+d)c)$$

the map we get from ψ_i, then we see by the same reasoning as above that the section corresponding to $\psi_i(z)$ is $\chi_i(p^*z) = e_{p^*z}(\chi_i) = p^*e_z(\tau_i)$. Hence we conclude that the image of $\varphi(z)$ in $\Gamma(Y_n, \mathcal{O}_{Y_n}(n-(1+d)c))$ is equal to the image of $\sum a_i\psi_i(z)$. Since $n-(1+d)c \geq q_0$ we have $\Gamma(Y_n, \mathcal{O}_{Y_n}(n-(1+d)c)) = I^{n-(1+d)c}/I^{2n-(1+d)c}$ by Lemma lemma-bound-q-and-d and we conclude the desired compatibility is true.

The case $d = 1$. Here we argue as above that we get

$$\chi : p^*M \longrightarrow \mathcal{O}_{Y_n}(n), \quad \sigma \in \Gamma(X, (p^*M)^\vee(n)), \quad \sigma' \in \Gamma(X, p^*(M^\vee)(n-2c)),$$

and then we use Lemma lemma-vanishing-coh-almost-projective part (4) to see that σ' is the image of some element $\tau \in I^{n-2c}M^\vee$. The rest of the argument is the same.

The case $d = 0$. Argument is exactly the same as in the case $d = 1$. □